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LITERATURE REVIEW: PRIVACY-PRESERVING VIRTUAL FITTING: A SYSTEMATIC LITERATURE REVIEW OF POSE ESTIMATION AND CLIENT-SIDE COMPUTER VISION FOR CLOTHING SIZE SELECTION

MSc Computing

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Abstract: Online fashion retail faces a persistent challenge of high return rates, primarily driven by uncertainty in clothing fit and size selection. This literature review examines key approaches to addressing this problem, with a focus on virtual try-on systems, pose estimation, and client-side computer vision. A systematic search was conducted across Scopus, ACM Digital Library, IEEE Xplore, SpringerLink, and Google Scholar, applying iterative refinement and defined selection criteria. The review shows that deep learning-based virtual try-on systems achieve high visual realism but typically rely on computationally intensive, server-side processing, raising concerns around scalability and privacy. In contrast, recent advances in pose estimation enable real-time body landmark detection directly in the browser, supporting lightweight and privacy-preserving alternatives. However, limited research has evaluated the effectiveness of such systems in improving user confidence in size selection. This review identifies a gap at the intersection of pose estimation, client-side deployment, and user-centred evaluation, and supports the development of a browser-based virtual fitting assistant aimed at improving sizing decisions in online retail.

Keywords: virtual fitting, virtual try-on, pose estimation, computer vision, client-side machine learning, edge computing, user trust, technology acceptance, clothing size prediction

1 Introduction

Online fashion retail has grown rapidly over the past decade, yet continues to suffer from a well-documented structural problem where customers cannot physically try on garments before purchase. This limitation drives return rates estimated at up to 40% of all online fashion orders, with incorrect size and poor fit consistently cited as the primary cause by Petersen and Kumar (2015). The financial and environmental costs of these returns arise from reverse logistics, re-packaging, inventory disruption, and excess carbon emissions. These factors have motivated substantial research and in-

dustry investment in technologies that can close the gap between the digital product image and the customer's lived experience of fit. A range of solutions have been proposed. At the data-driven end, researchers have developed machine learning models that infer likely fit from purchase history, body measurements, and customer review text. At the computer vision end, researchers have built virtual try-on systems that overlay digital clothing images onto photographs of real people. Most recently, advances in pose estimation and browser-based machine learning have opened a third possibility: lightweight, client-side systems that detect

body landmarks in real time from a webcam feed, derive relative body proportions, and use these to guide size selection, all without transmitting any images to a server.

This literature review supports a project proposal for a privacy-preserving virtual fitting assistant implemented as a browser-based web application. The proposed system uses MediaPipe Pose to extract body landmarks client-side, maps derived proportions to clothing size recommendations through a rule-based logic layer, and is evaluated through a user study measuring size selection confidence and usability. The review therefore focuses on four interconnected bodies of literature: (1) the problem of fit uncertainty and its consequences; (2) virtual try-on systems and their limitations as background context; (3) pose estimation frameworks and their suitability for browser-based deployment; and (4) privacy, edge computing, and user trust, the design considerations that motivate the proposed client-side architecture and that will shape the evaluation.

This topic represents a strong research area for three reasons. First, the practical problem is commercially urgent and well evidenced: sizeable return rates impose real costs and the literature is rich enough to justify a novel approach. Second, the technical problem is genuinely under-explored at the intersection of browser-based computer vision, privacy-by-design, and size recommendation which existing systems either sacrifice privacy for accuracy (cloud-based 3D scanning) or sacrifice practicality for privacy (requiring explicit body measurement entry). Third, the proposed approach is technically feasible within a short project timeframe because it applies and adapts existing pre-trained models rather than training new ones, making it a realistic and well-scoped Masters project.

2 Methodology

This article employs the systematic literature review by applying the following steps:

1. Definition of research questions.
2. Iterative search process.
3. Search results filtering criteria.
4. Quality assessment.

2.1 Research Questions

The literature review is guided by a set of research questions designed to structure the search, selection, and analysis of relevant studies. These questions reflect both the technical and user-centered aspects of the proposed virtual fitting assistant. They are as follows:

- **RQ1:** What approaches are currently used in virtual fitting and clothing size prediction systems?
- **RQ2:** How is pose estimation applied within virtual try-on and fitting applications?
- **RQ3:** What are the limitations of existing virtual fitting solutions, particularly in terms of scalability, accuracy, and privacy?
- **RQ4:** How are user trust, confidence, and acceptance evaluated in virtual fitting and related systems?

These questions ensure that the review captures not only the technical foundations of virtual fitting systems but also their practical implications for real-world use, particularly in supporting user decision-making in online fashion retail.

2.2 Databases Selected

Five databases were selected to ensure comprehensive and balanced coverage of both the technical and user-oriented aspects of the research topic.

Scopus¹ was used as the primary database due to its broad interdisciplinary coverage across computer science, human-computer interaction (HCI), and consumer behaviour research. This made it particularly suitable for identifying studies on virtual fitting systems, usability, and user perception.

ACM Digital Library² was selected for its strong focus on computing research, including conference proceedings such as CHI and RecSys. It is particularly valuable for identifying system implementations, user studies, and practical applications of virtual fitting and pose estimation technologies.

IEEE Xplore³ was included to capture core computer vision and pose estimation research, particularly from major venues such as CVPR, ICCV,

¹<https://www.scopus.com/pages/home#basic>

²<https://dl.acm.org/>

³<https://www.ieee.org/>

and ECCV, where foundational work such as pose estimation frameworks is typically published.

SpringerLink⁴ was used to identify applied research in machine learning and virtual fitting systems, including studies that bridge technical development and practical deployment contexts.

Finally, **Google Scholar**⁵ was used as a supplementary database to identify literature on technology acceptance, user trust, and consumer behaviour. These aspects are less extensively covered in technical computing databases but are essential for grounding the evaluation methodology of the proposed system.

In addition, preprint repositories such as **arXiv**⁶ were consulted selectively to identify recent developments in browser-based and client-side computer vision. However, only peer-reviewed sources were included in the final paper set.

2.3 Search Strategy and Iterative Process

A systematic and iterative search strategy was employed across all databases. Initial broad queries were used to capture the general research area, followed by progressive refinement using more specific technical keywords. This process ensured that result sets were reduced to manageable sizes (≈ 30 or fewer per database) while maintaining relevance to the research topic. The search process focused on three core themes:

1. Virtual try-on and virtual fitting systems
2. Pose estimation and body landmark detection
3. Privacy-preserving, client-side, and edge-based computer vision

Boolean operators (AND, OR) and grouping with parentheses were used to control query logic and avoid overly broad or ambiguous results. Filters were applied consistently across databases where possible, including publication year (2017–2026), English language, and peer-reviewed conference or journal papers.

2.4 Article Selection Criteria

To ensure the relevance and quality of the selected literature, a set of inclusion and exclusion criteria

was defined and systematically applied during the screening process.

2.4.1 Inclusion Criteria

Studies were included if they met the following conditions where applicable:

- Focus on **virtual try-on**, **virtual fitting**, or **clothing size prediction**
- Apply or evaluate **pose estimation** or **body landmark detection**
- Address **user trust**, **confidence**, or **technology acceptance** in interactive systems
- Present **empirical evaluation**, **system implementation**, or experimental results
- Published in **peer-reviewed journals** or **conference proceedings**
- Published between **2017 and 2026**
- Written in **English**

2.4.2 Exclusion Criteria

Studies were excluded if they met any of the following exclusion criteria(EC):

- **EC1:** Not directly related to clothing fit, virtual fitting, or pose estimation
- **EC2:** Focus solely on **3D garment simulation** without relevance to size prediction or user decision-making
- **EC3:** Concern **general computer vision tasks** (e.g., action recognition, medical pose analysis) without application to fitting or clothing
- **EC4:** Non-academic sources (e.g., blogs, articles without peer review)
- **EC5:** Duplicate studies retrieved from multiple databases
- **EC6:** Papers without sufficient methodological detail or evaluation

⁴<https://link.springer.com/>

⁵<https://scholar.google.com/>

⁶<https://arxiv.org/>

3 Search Results

3.1 Systematic Search Log

A combined search log is presented to provide a clear and traceable overview of the iterative search process across all selected databases in the Table 3.1.

3.2 Observations from the Search Process

The iterative refinement process revealed several important characteristics of the literature.

First, initial searches across all databases produced large result sets due to the broad use of terms such as “fashion” and “virtual try-on.” These terms often retrieved studies unrelated to sizing or fit, including general recommendation systems and image synthesis research. The use of more specific technical constraints, such as “computer vision”

and “pose estimation,” was therefore essential to improve relevance.

Second, the introduction of pose estimation as a filtering criterion significantly reduced result sizes while increasing alignment with the proposed system. In both Scopus and ACM Digital Library, this step reduced results to fewer than 50 papers, indicating that the intersection of pose estimation and virtual fitting remains a relatively focused research area.

Third, the search identified a relative scarcity of studies explicitly addressing privacy-preserving or client-side virtual fitting systems. While edge computing and on-device machine learning are well studied in general, their application to clothing fit and virtual fitting remains limited. This observation directly supports the research gap addressed by the proposed project.

Finally, differences between databases were observed. Scopus provided broad interdisciplinary coverage, ACM Digital Library contributed system-

Database	Search ID	Query Focus / Keywords	Results	Notes / Decision
Scopus	S1	(“virtual try-on” OR “virtual fitting”) AND (clothing OR fashion)	667	Initial broad search; high number of results due to general terms
	S2	S1 + (“computer vision” OR “deep learning”)	184	Improved relevance by focusing on computational approaches
	S3	S2 + (“pose estimation” OR “body landmark”)	29	Highly relevant; final Scopus shortlist used for screening
ACM DL	A1	(“virtual try-on” OR “virtual fitting”) AND (clothing OR fashion)	204	Broad but controlled baseline after applying date filters
	A2	A1 + (“computer vision” OR “deep learning”)	141	Reduced non-technical results
	A3	A2 + (“pose estimation” OR “body landmark”)	32	Highly relevant; used for selecting system-focused papers
IEEE Explore	I1	(“pose estimation” AND clothing AND fitting)	64	Initial search focused on pose estimation in clothing context
	I2	I1 + (“edge” OR “client-side” OR “browser”)	6	Refined to capture deployment-focused studies
Springer	SP1	(“virtual fitting” AND clothing)	50	Broad applied research coverage
	SP2	SP1 + (“pose estimation” OR “measurement”)	19	Focused on body measurement and fitting logic
Google Scholar	G1	(“virtual fitting” AND trust AND user)	21	Captures user perception and trust studies
	G2	G1 + (“technology acceptance” OR TAM)	11	Focused on evaluation frameworks for user studies

Table 3.1: Systematic iterative search log by database.

level and implementation-focused research, IEEE Xplore emphasised foundational computer vision methods, and Google Scholar provided valuable insights into user trust and technology acceptance. Together, these sources ensured a balanced and comprehensive literature base.

After applying the inclusion and exclusion criteria, a total of 19 papers were selected for detailed review. These papers represent a balanced set of studies across virtual try-on systems, pose estimation, body measurement, and user-centred evaluation, and form the basis of the analysis presented in Chapter 4.

4 Literature Review

4.1 Fit Uncertainty and the Return Problem

A central challenge in online fashion retail is the inability of customers to physically assess garment fit prior to purchase. This contributes to uncertainty in size selection and increases the likelihood of product returns. Although Petersen and Kumar (2015) falls outside the main 2017–2026 search range, it is retained as a foundational study linking perceived risk and return behaviour in retail contexts.

Within the selected literature, virtual fitting technologies are increasingly proposed as a solution. The literature shows growing interest in AI-supported e-commerce fitting tools, including mobile virtual fitting applications, recommendation systems, and real-time try-on systems (Shakeel et al., 2025; Srinivasan et al., 2025; Deshmukh and Pai, 2025). These studies indicate that virtual fitting is increasingly viewed not only as a visualisation tool, but also as a decision-support mechanism for online shopping.

However, research on garment fit highlights that accurately matching clothing to diverse body shapes remains complex, as fit depends on both objective measurements and subjective perception (Liu et al., 2022). This indicates that addressing fit uncertainty requires both technical modelling and user-centred considerations.

4.2 Virtual Try-On Systems: The State of the Art and Its Limitations

Virtual try-on systems form the largest body of literature identified in both Scopus and ACM Digital Library. These systems commonly use computer vision, deep learning, image synthesis, garment warping, pose estimation, and segmentation to generate images of users wearing target garments. For example, DL-VTON combines OpenPose, segmentation, garment warping, image inpainting, and transformer-based blending to generate image-based try-on outputs (Yanda et al., 2025). Similarly, CloTH-VTON+ explores hybrid image-based and 3D reconstruction approaches for clothing virtual try-on (Minar et al., 2021).

ACM studies demonstrate continued progress toward higher visual fidelity. FashionOn uses semantic guidance and pose information to preserve fine garment details and handle occlusions (Hsieh et al., 2019), while M2E-Try On Net enables clothing transfer directly from model images without requiring clean product inputs (Wu et al., 2019). More recent approaches, such as Shape-Guided Clothing Warping for Virtual Try-On, focus on improving garment-body alignment and reducing distortions, particularly in regions involving complex pose or exposed limbs (Han et al., 2024).

In contrast, IEEE research places greater emphasis on system adaptability and deployment. Systems such as Tryly (Nikkitha et al., 2025) and limb-aware virtual try-on networks (Zhang et al., 2024) incorporate augmented reality and adaptive warping techniques to improve interaction and robustness in practical environments.

Similarly, broader survey work from Springer-Link highlights the evolution of virtual try-on systems from simple image overlays to complex pipelines combining deep learning, garment modelling, and human reconstruction (Ghodhbani et al., 2022; Fung et al., 2026).

Despite these advances, key limitations persist. Most virtual try-on systems prioritise visual realism over accurate size prediction, limiting their effectiveness as decision-support tools. Furthermore, many approaches rely on computationally intensive, server-side pipelines, which restrict scalability and reduce suitability for lightweight, browser-based deployment.

This reveals a clear gap between visually realistic systems and practical, accessible solutions for everyday online shopping. This disconnect between visual realism and decision-support functionality highlights a key limitation in current virtual fitting research.

4.3 Pose Estimation: From Server to Lightweight Systems

Pose estimation is a key enabling technology across the reviewed literature. Several systems integrate pose estimation and body modelling to support virtual fitting. For example, SyncFit combines MediaPipe Pose, ARCore, and SMPL modelling for real-time mobile fitting assistance (Shakeel et al., 2025). Similarly, frameworks such as Online Virtual Try-On using Computer Vision (Tiwari et al., 2025) and Limb-Aware Virtual Try-On Network (Zhang et al., 2024) demonstrate how pose estimation supports garment alignment.

ACM research provides further insights. ARShoe demonstrates real-time augmented reality try-on on smartphones using pose estimation and segmentation (An et al., 2021). Zheng et al. (2022) propose a lightweight graph transformer network for reconstructing human mesh from 2D pose, addressing efficiency and deployment challenges. Poullos et al. (2024) further show how MediaPipe enables real-time 3D pose estimation using RGB input.

IEEE studies extend this by focusing on measurement extraction. Shah et al. (2019) demonstrate how body dimensions can be estimated directly from images, providing a structured basis for size recommendation.

SpringerLink research complements these approaches by exploring the extraction of body feature parameters using image analysis techniques as presented by Wei et al. (2018). These methods offer a bridge between pose estimation and size prediction, although they often rely on controlled conditions.

Overall, pose estimation enables lightweight and real-time body analysis; however, its direct application to practical clothing size decision support remains limited and insufficiently explored.

4.4 Client-Side Processing, Real-Time Deployment, and Privacy

A key motivation for the proposed system is the use of client-side processing to preserve user privacy and improve responsiveness. Several studies demonstrate the feasibility of real-time virtual fitting in mobile and embedded environments. For instance, ARShoe enables real-time augmented reality try-on on smartphones using pose estimation and segmentation (An et al., 2021), while SyncFit provides a mobile-based fitting application with interactive real-time feedback (Shakeel et al., 2025). Similarly, RGB-based pose estimation approaches demonstrate that real-time body tracking can be achieved using lightweight frameworks (Poullos et al., 2024).

However, privacy is rarely treated as a primary design objective within the existing literature. Most systems focus on improving visual realism, pose accuracy, or rendering performance, with limited consideration of how user data is processed or stored. This is particularly significant in virtual fitting contexts, where systems often require access to user images or body-related data.

As a result, client-side, browser-based pose estimation offers a promising direction for virtual fitting systems. It enables real-time body analysis while reducing the need to transmit sensitive images or body-related data to external servers. However, this deployment model remains relatively underexplored in clothing size decision support.

4.5 User Confidence and Acceptance in Sizing Tools

User perception plays a critical role in the adoption of virtual fitting technologies. The Technology Acceptance Model suggests that perceived usefulness and ease of use are key factors influencing user adoption (Davis et al., 1989; Venkatesh et al., 2003; Shen et al., 2021).

Studies on virtual fitting and augmented reality shopping indicate that user trust is closely linked to perceived accuracy, usability, and transparency. Systems such as Tryly and SyncFit incorporate user interaction and usability considerations (Shakeel et al., 2025), while Chong et al. (2021) include user studies to evaluate system performance.

Research on garment fit further highlights that

user satisfaction depends on subjective factors such as comfort, expectations, and perceived accuracy (Liu et al., 2022). Studies on AR and virtual fitting confirm the importance of trust and usability (Mendoza-Ramírez et al., 2023; Jim et al., 2023). This suggests that effective virtual fitting systems must provide accurate outputs while also clearly communicating results to support user confidence.

4.6 Synthesis and Research Gap

The reviewed literature across Scopus, ACM, IEEE, and SpringerLink demonstrates significant progress in virtual try-on systems, pose estimation, and real-time fitting applications. Virtual try-on research has advanced in visual realism through deep learning, garment warping, and 3D reconstruction (Hsieh et al., 2019; Minar et al., 2021; Yanda et al., 2025; Han et al., 2024). Pose estimation and body modelling techniques enable lightweight and real-time analysis (An et al., 2021; Zheng et al., 2022; Poullos et al., 2024; Shakeel et al., 2025). SpringerLink studies highlight body modelling and measurement extraction, while IEEE research provides insights into anthropometric modelling and system deployment.

However, these areas remain only partially integrated. Virtual try-on systems prioritise visual realism over practical size selection. Pose estimation methods focus on body representation but are rarely used directly for decision support. Real-time systems demonstrate feasibility, but privacy-preserving client-side solutions remain underexplored. Additionally, user trust and confidence are not consistently incorporated into evaluation. This fragmentation across research directions highlights a lack of integrated solutions that simultaneously address technical feasibility, privacy, and user trust.

Consequently, a clear research gap exists at the intersection of pose estimation, client-side processing, and user-centred evaluation. Existing systems either prioritise visual realism without supporting size decision-making, or focus on body representation without addressing usability and trust. There is limited research that integrates browser-based pose estimation with lightweight, privacy-preserving design while explicitly evaluating its impact on user confidence. Addressing this gap provides an opportunity to develop more practical, accessible, and trustworthy virtual fitting systems for

online fashion retail.

5 Conclusions

This literature review set out to systematically examine research on virtual fitting systems, pose estimation, and client-side computer vision in the context of clothing size selection. The search process demonstrated that while a large body of literature exists on virtual try-on and computer vision, relatively few studies directly address the problem of fit uncertainty using lightweight, privacy-preserving approaches.

The systematic search approach proved effective in identifying relevant literature across multiple databases. The use of iterative refinement, combined with clearly defined inclusion and exclusion criteria, allowed the initial large result sets to be reduced to a focused and manageable collection of high-quality studies. The process highlighted the importance of carefully selecting keywords and applying domain-specific constraints, particularly when working with broad terms such as “fashion” or “virtual try-on,” which often return irrelevant results.

Several key insights emerged from the review. First, virtual try-on systems have made significant progress in visual realism, but they remain computationally intensive and typically rely on server-side processing. Second, pose estimation techniques offer a promising alternative for lightweight, real-time body analysis, particularly in browser-based environments. Third, while privacy and edge computing are increasingly important considerations, their application to virtual fitting remains limited. Finally, user trust and confidence are critical factors influencing adoption, yet they are often insufficiently integrated into system design and evaluation.

5.1 Limitations of the Search

Despite the structured approach, several limitations were identified. The use of predefined keywords may have excluded relevant studies that use alternative terminology. Differences in indexing and search behaviour across databases also affected the consistency of results. For example, Google Scholar returned broader result sets compared to more con-

trolled databases such as Scopus, requiring additional manual filtering.

The selected publication range (2017–2026) prioritised recent developments; however, a small number of earlier foundational studies were also included where necessary. Additionally, access limitations to certain databases (e.g. Web of Science) influenced the selection of sources, although alternative databases were used to mitigate this constraint.

5.2 Reflections on the Search Process

Overall, the systematic literature search was effective in identifying a relevant and balanced set of papers. The iterative refinement process was particularly valuable in improving the precision of search results, while the use of multiple databases ensured coverage across both technical and user-focused research domains.

The process also revealed the importance of aligning the search strategy with the research objectives. In particular, the addition of privacy- and deployment-related keywords helped uncover a gap in the literature that may not have been visible through a purely technical search. This demonstrates the value of combining domain knowledge with systematic search techniques when conducting literature reviews in computing.

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